



**Faculty of Mechanical
Engineering and Robotics**

**Department of Robotics and
Mechatronics**



Signal processing and identification in control of mechatronic devices

Topic: Introduction to image processing in MATLAB

Goal: Students are introduced to working with matrix data and images in MATLAB programming environment. Students learn how to analyze image histograms.

Problems:

1. Basic function for image processing available in MATLAB programming environment,
2. Image histograms and histogram equalization,

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1. Basic functions for image processing in MATLAB programming environment.

Useful commands for beginning of each m-file: `cls`, `clear all`, `close all`,

Loading of an image: `A = imread('file path');`

Write file path to the image or name of the image with file extension if you're working in a directory in which you store the image (working directory in MATLAB).

Computation of image resolution and number of color channels of loaded image.

`im_size = size(A);`

What is the meaning of numbers returned by the function?

Displaying an image: `imshow(A);`

Conversion of color image in grayscale image: `A_gray = rgb2gray(A);`

How will the vector returned by `size()` function change?

Saving modified image on disk:

`imwrite(A_gray, 'file path');`

Again, write file path to the image or name of the image with file extension if you're working in a directory in which you store the image (working directory in MATLAB).

Cutting subimages using indices:

Give range of rows and columns in image matrix, which will be copied to new image matrix, for example: `B = A(20:100, 50:150);`

Remember that first index means the number of **matrix row** and the second one means the number of **matrix column**. In MATLAB, the origin of image coordinate frame is at the top left image corner. Therefore, indices are incremented from **top to bottom** and from **left to right**.

In order to read pixel coordinates (and value of the pixel) use function:

`[x y value] = impxel(A)`

Point pixel of interest and click LMB. When you are finished with choosing pixels, press enter.

Function which cuts subimage:

- 1) Choosing subimage to cut manually. Point at the top left corner of subimage to be cut, mark the rectangular area to be cut, click RMB on the rectangle's side and chose *crop image* from the menu which will appear.

`[A_small rectangle] = imcrop(A);`

- 2) Automatic cutting of a subimage, when the size of the subimage and its location is known. Numbers x and y mean the coordinates of top left subimage corner and parameters width and height mean size of rectangle's sides.

```
A_small = imcrop(A, [x y width height]);
```

Computation of image histogram.

In order to compute and visualize image histogram use `imhist(A)` function. Function's argument is matrix representing grayscale image (it is not RGB image!).

What is the meaning of a histogram? What is its shape for different types of images (dark, bright, low and high contrast)?

Histogram equalization.

In order to equalize histogram of an image (if you want to obtain image with better contrast), you need to use function `histeq(A)`. The argument of the function is image in grayscale.

Exercises:

1. Load image from a set available on classes. Make small images out of image's fragments. Display histograms of these images. How they differ from each other?
2. Generate random grayscale image using different random number generators available in MATLAB. Display histograms of these images. Could you predict the shape of histograms? Random numbers drawn from different distributions could be generated using Statistics Toolbox.
3. Load images and perform image equalization on all of them. Display their histograms before and after image equalization operation. What is the difference?